Exercise - 3.12

Choose the correct answer or the most suitable answer:

1. $\frac{1}{\cos 80^\circ} - \frac{\sqrt{3}}{\sin 80^\circ} =$

   (1) $\sqrt{2}$ (2) $\sqrt{3}$ (3) 2 (4) 4

2. If $\cos 28^\circ + \sin 28^\circ = k^3$, then $\cos 17^\circ$ is equal to

   (1) $\frac{k^3}{\sqrt{2}}$ (2) $-\frac{k^3}{\sqrt{2}}$ (3) $\pm\frac{k^3}{\sqrt{2}}$ (4) $-\frac{k^3}{\sqrt{3}}$

3. The maximum value of $4\sin^2 x + 3\cos^2 x + \sin\frac{x}{2} + \cos\frac{x}{2}$ is

   (1) $4+\sqrt{2}$ (2) $3+\sqrt{2}$ (3) 9 (4) 4

4. $\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right) =$

   (1) $\frac{1}{8}$ (2) $\frac{1}{2}$ (3) $\frac{1}{\sqrt{3}}$ (4) $\frac{1}{\sqrt{2}}$

5. If $\pi < 2\theta < \frac{3\pi}{2}$, then $\sqrt{2+\sqrt{2+2\cos 4\theta}}$ equals to

   (1) $-2\cos\theta$ (2) $-2\sin\theta$ (3) $2\cos\theta$ (4) $2\sin\theta$

6. If $\tan 40^\circ = \lambda$, then $\frac{\tan 140^\circ - \tan 130^\circ}{1+\tan 140^\circ \tan 130^\circ} =$

   (1) $\frac{1-\lambda^2}{\lambda}$ (2) $\frac{1+\lambda^2}{\lambda}$ (3) $\frac{1+\lambda^2}{2\lambda}$ (4) $\frac{1-\lambda^2}{2\lambda}$

7. $\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \ldots + \cos 179^\circ =$

   (1) 0 (2) 1 (3) -1 (4) 89

8. Let $f_k(x) = \frac{1}{k}\left[\sin^k x + \cos^k x\right]$ where $x \in R$ and $k \geq 1$. Then $f_4(x) - f_6(x) =$

   (1) $\frac{1}{4}$ (2) $\frac{1}{12}$ (3) $\frac{1}{6}$ (4) $\frac{1}{3}$

9. Which of the following is not true?

   (1) $\sin\theta = -\frac{3}{4}$ (2) $\cos\theta = -1$ (3) $\tan\theta = 25$ (4) $\sec\theta = \frac{1}{4}$

10. $\cos 2\theta \cos 2\phi + \sin^2(\theta - \phi) - \sin^2(\theta + \phi)$ is equal to

    (1) $\sin 2(\theta+\phi)$ (2) $\cos 2(\theta+\phi)$ (3) $\sin 2(\theta-\phi)$ (4) $\cos 2(\theta-\phi)$

11. $\frac{\sin(A-B)}{\cos A\cos B} + \frac{\sin(B-C)}{\cos B\cos C} + \frac{\sin(C-A)}{\cos C\cos A}$ is

    (1) $\sin A + \sin B + \sin C$ (2) 1 (3) 0 (4) $\cos A + \cos B + \cos C$

12. If $\cos p\theta + \cos q\theta = 0$ and if $p \neq q$, then θ is equal to (n is any integer)

    (1) $\frac{\pi(3n+1)}{p-q}$ (2) $\frac{\pi(2n+1)}{p\pm q}$ (3) $\frac{\pi(n\pm 1)}{p\pm q}$ (4) $\frac{\pi(n+2)}{p+q}$